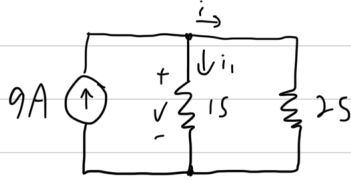


2.33

由于 $6S$ 元件与 $3S$ 元件串联, 可视为 $\frac{6 \times 3}{6+3} = 2S$ 元件, 其与 $2S$ 元件并联视为 $4S$ 元件, 又与 $4S$ 元件串联, 可视为 $\frac{4 \times 4}{4+4} = 2S$ 元件

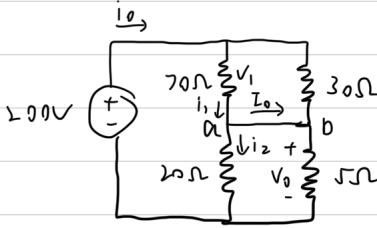
电路可被等效为:



由分流规律可知 $i = \frac{1}{1+\frac{1}{2}} \times 9A = 6A$

$i_1 = 9 - i = 3A$, $V = i_1 \cdot \frac{1}{1} = 3V$

2.35



由 $70 \parallel 30 = \frac{70 \times 30}{70+30} = 21\Omega$, $20 \parallel 5 = \frac{20 \times 5}{20+5} = 4\Omega$

$i_0 = \frac{200}{21+4} = 8A$

$V_1 = 21i_0 = 168V$, $V_0 = 4i_0 = 32V$

$i_1 = \frac{V_1}{30} = 2.4A$, $i_2 = \frac{V_0}{5} = 1.6A$

由 KCL, 对于 α 点 $i_1 = i_2 + I_0$, 即 $I_0 = 2.4 - 1.6 = 0.8A$

则 $V_0 = 32V$, $I_0 = 0.8A$

2.41

设 R_1 为三个 12Ω 并联的等效电阻, $\frac{1}{R_1} = \frac{1}{12} + \frac{1}{12} + \frac{1}{12}$ 即 $R_1 = 4\Omega$

则 $R_{eq} = 30 + 60 \parallel (10 + R_1 + R_0) = 30 + 60 \parallel (14 + R)$

即 $50 = 30 + \frac{60 \times (14 + R)}{74 + R}$, 即 $74 + R = 42 + 3R$, 则 $R = 16\Omega$

2.74

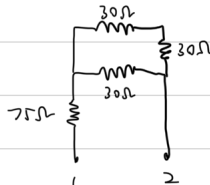
在高档时, $6 = (0.01 + R_1 + 0.02) \times 5$, 可得 $R_1 = 1.17\Omega$

在中档时, $6 = (0.01 + R_2 + R_1 + 0.02) \times 3$ 可得 $R_2 + R_1 = 1.97$, $R_2 = 1.97 - 1.17 = 0.8\Omega$

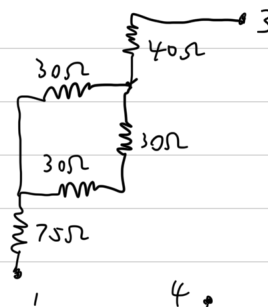
在低档时, $6 = (0.01 + R_3 + R_2 + R_1 + 0.02) \times 1$ 可得 $R_1 + R_2 + R_3 = 5.97$, $R_1 = 5.97 - 1.97 = 4\Omega$

2.82

(a) $R_{12} = 75 + \frac{60 \times 30}{60+30} = 95\Omega$



(b) $R_{13} = 40 + 75 + \frac{60 \times 30}{60+30} = 135\Omega$



(c) $R_{14} = 40 + 75 = 115\Omega$

